

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE CODE: CSA02 COURSE NAME: C Programming

COURSE OUTCOME COVERED

CO1: Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks. (BL2, BL3)

Assessment Tool Weightage: 40%

QUESTIONS: Total Marks:100 Annexure A

PSEUDO CODE WRITING

Code of Conduct:

I A.Guna Sekhar/192525302

(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

A Guna Sekar
Signature of the student:

Annexure A

PSEUDO CODE WRITING

Q1. Given pseudocode:

```
#define DEBUG 0
```

```
IF DEBUG
```

```
PRINT intermediate_value
```

```
END IF
```

- Identify error in directive usage
- Correct the logic

Q2. #define PI _____

```
DECLARE radius AS float
```

```
DECLARE area AS float
```

```
INPUT radius
```

```
area = _____
```

```
PRINT area
```

Fill:

- Constant
- Formula

Q3. #define RATE 1.5

```
DECLARE x AS int = 10
```

```
DECLARE result AS float
```

```
result = x * RATE
```

```
PRINT result
```

Questions:

- Output?
- Why is the result float?

Q4. #define MAX 1000

```
DECLARE temp AS int
```


INPUT temp

IF temp > MAX

PRINT "Valid"

ELSE

PRINT "Invalid"

- Identify logical error
- Suggest correct condition

Q5. Given unordered pseudocode steps:

- DEFINE conversion rate
- PRINT result
- INPUT amount
- CALCULATE converted value

Arrange in correct order

Q6. #define MIN 10

#define MAX 50

DECLARE value AS int

INPUT value

_____ (value >= MIN AND value <= MAX)
PRINT "Within range"

PRINT "Out of range"

Fill missing control structures

Q7. A billing system includes optional features such as discount calculation and tax computation. Depending on the business requirement, these features may be enabled or disabled at compile time. The program should calculate the total bill amount based on input price and quantity. If discount and tax features are enabled, they should be applied using predefined constants; otherwise, they should be skipped. The system must ensure that floating-point precision is maintained during calculations.

Write pseudocode using preprocessor directives to enable/disable features, define constants for discount and tax rates, and use appropriate data types.

Q8. #define TAX 5%


```
DECLARE price AS int
DECLARE total AS int
```

```
total = price + (price * TAX)
```

```
PRINT total
```

- Identify errors
- Correct data types and constant definition

9. A temperature conversion program supports multiple modes such as Celsius to Fahrenheit and Fahrenheit to Celsius. The mode of operation should be selected at compile time using preprocessor directives. The program should use appropriate floating-point data types to ensure accuracy and define conversion formulas using symbolic constants wherever applicable. The final output should be displayed with two decimal precision.

Write pseudocode using preprocessor directives for mode selection, constants for conversion formulas, and proper data types.

10. #define SIZE 5

```
DECLARE arr[SIZE]
INPUT elements
```

- Extend pseudocode to:
 - Find sum
 - Compute average

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficie nt	Level 2 – Developi ng	Level 1 – Beginnin g
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understa nd ing	Incorrect interpreta tio n

Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

C-programming

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Course Code: - CSA0203

Q1) Error in directive

Pseudocode:

```
#define DEBUG 1
```

```
IF DEBUG == 1
```

```
    PRINT intermediate-value
```

```
END IF
```

Q2) Area of circle

```
#define PI 3.14
```

```
DECLARE radius AS float
```

```
DECLARE area AS float
```

```
INPUT radius
```

```
area = PI * radius * radius
```

```
print area
```

Q3

```
#define RATE 15
```

```
DECL x AS int = 10
```

```
DECLARE result AS float
```

```
result = x * RATE
```

```
PRINT result
```


Q4)

```
#define max 1000  
DECLARE temp AS int  
INPUT temp  
IF temp <= MAX  
    print "Valid"  
ELSE  
    PRINT "INVALID"  
END IF
```

Q5)

```
DEFINE Conversion rate  
INPUT amount  
CALCULATE converted value  
PRINT result
```

Q6)

```
#define MIN 10  
#define MAX 50  
DECLARE value AS int  
INPUT value  
IF (value >= MIN AND value <= MAX)  
    PRINT "Within range"  
ELSE  
    PRINT "out of range"  
END IF
```



```
#define DISCOUNT_ENABLED 1
#define TAX_ENABLED 1
```

```
#define DISCOUNT_RATE 0.10
#define TAX_RATE 0.05
```

```
DECLARE price AS float
DECLARE quantity AS int
DECLARE total AS float
```

```
INPUT price
```

```
INPUT quantity
```

```
total = price * quantity
```

```
IF DISCOUNT_ENABLED == 1
```

```
    total = total - (total * DISCOUNT_RATE)
```

```
END IF
```

```
IF TAX_ENABLED == 1
```

```
    total = total + (total * TAX_RATE)
```

```
END IF
```

```
PRINT total
```

Q8)

```
#define TAX 0.05
```

```
DECLARE price AS float
```

```
DECLARE total AS float
```

```
INPUT price
```

```
total = price + (price * TAX)
```

```
print total
```


Q9) #define MODE 1 // 1 = C to F, 2 = F to C

DECLARE temp AS float

DECLARE result AS float

INPUT temp

IF MODE == 1 THEN

result = (temp * 9/5) + 32

ELSE

result = (temp - 32) * 5/9

END IF

PRINT result

Q10)

#define SIZE 5

DECLARE arr [SIZE]

DECLARE sum AS int = 0

DECLARE avg AS float

FOR i = 0 TO SIZE - 1

INPUT arr[i]

sum = sum + arr[i]

END FOR

avg = sum / SIZE

PRINT sum

PRINT avg